

EDUGENIE

AI-Based Educational Platform

PROJECT DEVELOPMENT PHASE

Project Development Document

Educational Technology Project

1. Introduction

The Project Development Phase is the stage where the planned design and requirements of EduGenie are converted into a working application. In this phase, developers build the system, integrate AI features, connect the database, and test each module.

2. Development Objectives

- To develop the EduGenie application according to the requirements.
- To implement the designed features and functions.
- To integrate AI-based learning assistance.
- To connect and manage the database.
- To create a simple and user-friendly interface.

3. Frontend Development

The frontend is developed to provide an easy-to-use interface for students.

Main features include:

- Login and registration.
- Student dashboard.
- Study materials section.
- Quiz and assessment interface.
- Progress tracking.
- AI chatbot or learning assistant.

4. Backend Development

The backend manages the main operations of EduGenie.

It includes:

- User authentication.
- Student profile management.
- Processing user requests.
- Managing study materials.
- Quiz and score management.
- Communication between frontend and database.

5. Database Development

A database is used to store and manage important information such as:

- Student details.
- Login information.
- Study materials.
- Quiz questions and answers.
- Scores and progress.
- Learning history.

6. AI Integration

AI is integrated into EduGenie to provide personalized learning support.

It can be used for:

- Answering student questions.
- Explaining difficult concepts.
- Generating study suggestions.
- Providing personalized learning recommendations.
- Supporting quiz and practice activities.

7. Module Development

The system can be developed using separate modules:

1. User Management Module
2. Learning Material Module
3. AI Assistant Module
4. Quiz and Assessment Module
5. Progress Tracking Module
6. Database Management Module

8. Testing During Development

Each module is tested during development to identify and fix errors.

Testing includes:

- Unit testing.
- Integration testing.
- Functional testing.
- User interface testing.
- Database testing.

9. Deployment

After successful development and testing, EduGenie is prepared for deployment. The application is hosted on the required server or cloud platform so that students can access it.

10. Expected Outcome

The expected outcome is a functional AI-based educational platform that provides students with learning resources, personalized assistance, quizzes, and progress tracking.

11. Conclusion

The Project Development Phase converts the EduGenie project plan and design into a working system. By developing each module, integrating AI and database features, and performing proper testing, EduGenie can provide an effective and user-friendly learning platform.